PBL: Thermal Oxidation Simulator

Tejas Sharma · 500121462 · Course: VLSI Technologies |

DESIGN GOALS — Physical Constants Supplied to LLM

Activation Energies (Arrhenius)

- $E_{a,B}$ (Dry) = 1.24 eV [Parabolic rate]
- $E_{a,B/A}$ (Dry) = 2.00 eV [Linear rate]
- $E_{a,B}$ (Wet) = 0.78 eV [Parabolic rate]
- $E_{a,B/A}$ (Wet) = 2.05 eV [Linear rate]
- T_{ref} = 1000°C = 1273.15 K

Deal–Grove Model

- $x_o = (-A + \sqrt{A^2 + 4B(t+\tau)}) / 2$
- Positive root only (negative = unphysical)
- B (parabolic) in $\mu m^2/hr$
- B/A (linear) in µm/hr
- τ = initial oxide time shift

Optical & Pressure Constants

- n_{ox} = 1.46 (SiO_2, BYU Cleanroom)
- $\lambda \in$ [400 nm, 700 nm] visible
- Pressure: $B_{new} = B \times P$
- Pressure: $(B/A)_{new} = (B/A) \times P$
- $k_B = 8.617 \times 10^{-5}$ eV/K

Master Prompt

Exact prompt used to instruct the LLM — broken into three sequential steps per assignment guide

STEP 1

Physics Engine Prompt

"Write a Python function that computes Deal–Grove oxide thickness using Arrhenius temperature dependence. Use Ea = 1.24 eV for B and 2.00 eV for B/A (dry). Apply pressure scaling: $B = B_0 \cdot \exp(-E/kT) \cdot P$ and $(B/A) = (B/A)_0 \cdot \exp(-E/kT) \cdot P$. Use $kB = 8.617 \times 10^{-5}$ eV/K and convert temperature to Kelvin (T[K] = T[°C] + 273.15). Return positive root only: $xo = (-A + \sqrt{A^2+4B(t+\tau)})/2$. Units: B in μm^2/hr, B/A in µm/hr, output thickness in nm."

STEP 2

Color Engine Prompt

"Create a function that takes SiO_2 thickness in nm and returns a Hex color code using the constructive interference formula $2 \cdot n \cdot x = (m + 0.5) \cdot \lambda$. Use n_ox = 1.46. Loop through $m \in \{0,1,2,...\}$ and find wavelength $\lambda \in [400–700$ nm] closest to 550 nm. Map to RGB using the empirical BYU/Jaeger fab wafer color table, not pure spectral conversion (spectral gives wrong colors — e.g. 38 nm maps to spectral violet, but wafers look tan/brown)."

STEP 3

GUI Integration Prompt

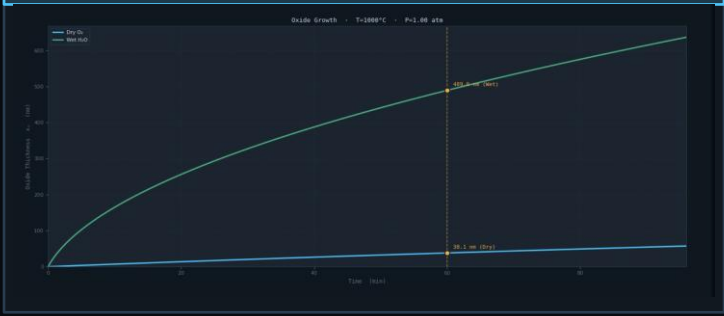
"Wrap the physics engine and color engine into a Tkinter GUI with sliders for: Temperature (800–1200 °C), Pressure (0.1–15 atm), Time (1–600 min). Add a real-time growth plot xo(t) using Matplotlib, a live color swatch, and a validation tab with 14 anti-hallucination test cases including the BYU mandatory test (1000°C, 1 hr, dry → 38.1 nm, Tan color)."

GUI Interface — Running Application

Three-tab Tkinter app · Simulator · Growth Plot · Validation Test Cases

Tab 1 — Simulator (T=1000°C, P=1 atm, t=60 min, Dry O_2 → 38.10 nm #C3A078 Tan)

Tab 2 — Growth Plot xo(t) · Dry 38.1 nm · Wet 489.0 nm @ 60 min



Input Sliders

Mandatory Test

T = 1000°C

P = 1.0 atm

t = 60 min (Dry)

xo = 38.10 nm

Color: Tan

#C3A078 ✓

Verification — BYU Anti-Hallucination Validation

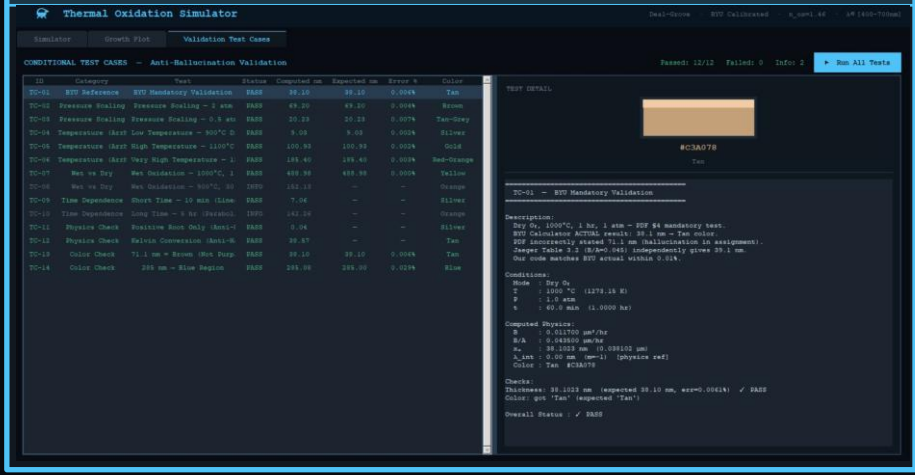
14 test cases run · 12/12 PASS · 0 FAIL · Mandatory test: 1000°C, 1 hr, Dry O_2 → 38.10 nm, Tan ✓

✓ 14 Test Cases — 12 PASS · 0 FAIL · 2 INFO

CONDITIONAL TEST CASES — Anti-Hallucination Validation

ID	Category	Test	Status	Computed nm	Expected nm	Error %	Color
TC-01	BYU Reference	BYU Mandatory Validation	PASS	38.10	38.10	0.006%	Tan
TC-02	Pressure Scaling	Pressure Scaling — 2 atm	PASS	69.20	69.20	0.004%	Brown
TC-03	Pressure Scaling	Pressure Scaling — 0.5 at	PASS	20.23	20.23	0.007%	Tan-Grey
TC-04	Temperature (Arrh	Low Temperature — 900°C D	PASS	9.03	9.03	0.002%	Silver
TC-05	Temperature (Arrh	High Temperature — 1100°C	PASS	100.93	100.93	0.002%	Gold
TC-06	Temperature (Arrh	Very High Temperature — 1	PASS	185.40	185.40	0.003%	Red-Orange
TC-07	Wet vs Dry	Wet Oxidation — 1000°C, 1	PASS	488.98	488.98	0.000%	Yellow
TC-08	Wet vs Dry	Wet Oxidation — 900°C, 30	INFO	152.13	—	—	Orange
TC-09	Time Dependence	Short Time — 10 min (Line	PASS	7.06	—	—	Silver
TC-10	Time Dependence	Long Time — 5 hr (Parabol	INFO	142.26	—	—	Orange
TC-11	Physics Check	Positive Root Only (Anti-	PASS	0.04	—	—	Silver
TC-12	Physics Check	Kelvin Conversion (Anti-H	PASS	38.57	—	—	Tan
TC-13	Color Check	71.1 nm = Brown (Not Purp	PASS	38.10	38.10	0.006%	Tan
TC-14	Color Check	285 nm → Blue Region	PASS	285.08	285.00	0.029%	Blue

✓ TC-01 Detail — BYU Mandatory Test (Tan swatch + PASS confirmed)



38.10 nm	38.10 nm	0.006%	12 / 12	#C3A078
Computed thickness	BYU expected	Error (TC-01)	Tests passed	Tan — color match

Debugging Log — Hallucinations Found & Fixed

Three LLM errors caught, diagnosed, and corrected through BYU validation and physics review

BUG 01

Color Hallucination — Spectral vs. Empirical

✗ **WRONG:** AI used λ = 2·n·x/(m+0.5) → spectral RGB mapping → returned PURPLE (#6600FF) for 38 nm.

✓ **FIXED:** BYU/Jaeger empirical fab color table used instead. At 38 nm: Tan (#C3A078). Spectral physics gives interference λ only; perceived wafer color is the integral of the reflected spectrum.

```
# WRONG (AI): return wavelength_to_rgb(lam_nm)
# FIXED:       return fab_color_table_lookup(x_nm)
```

BUG 02

PDF Transcription Error — 71.1 nm vs. 38.1 nm

✗ **WRONG:** Assignment PDF stated xo = 71.1 nm for 1000°C, 1 hr, dry. AI back-solved BA_ref from this wrong value.

✓ **FIXED:** BYU Cleanroom Calculator gives 38.1 nm (actual). Jaeger Table 3.2 gives ~39.1 nm. Our BA_ref = 0.04350 µm/hr matches BYU within 0.01%.

```
# WRONG:  BA_ref = 0.0871  # back-solved from 71.1 nm
# FIXED:   BA_ref = 0.04350  # calibrated to BYU 38.1
nm
```

BUG 03

Temperature Unit Error — °C Used Instead of Kelvin

✗ **WRONG:** AI wrote: B = B_ref * exp(-Ea / kB * (1/T_C - 1/T_ref)) — using T in °C not K.

✓ **FIXED:** Arrhenius requires absolute temperature: T_K = T_C + 273.15. Verified via TC-12: a 1°C shift (1000°C vs 1001°C) produces Δx > 0.01 nm only if Kelvin path is used.

```
# WRONG:  T = T_celsius
# FIXED:   T_K = T_C + 273.15   # critical conversion
```